

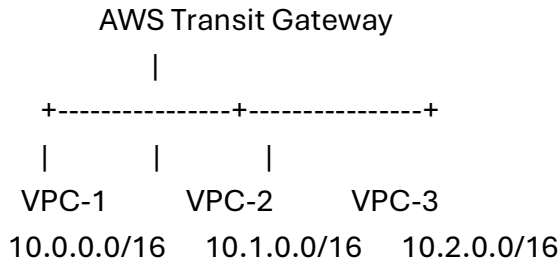
EOD Document – AWS Transit Gateway & VPC Endpoint

1. AWS Transit Gateway – Scalable Network Architecture

Objective

Design and deploy a scalable network architecture using **AWS Transit Gateway (TGW)** to simplify connectivity between multiple VPCs.

Architecture



Steps Performed

1. Created multiple VPCs with non-overlapping CIDR ranges.
 - a. VPC-1: 10.0.0.0/16
 - b. VPC-2: 10.1.0.0/16
 - c. VPC-3: 10.2.0.0/16
2. Created an **AWS Transit Gateway**.
3. Created Transit Gateway attachments for the VPCs.
4. Associated the VPC attachments with the Transit Gateway route table.
5. Added routes in the VPC route tables to send traffic for other VPC CIDRs through the Transit Gateway.
6. Verified connectivity between resources in the connected VPCs.

```
[ec2-user@ip-10-0-75-50 ~]$ ping 172.31.8.16
PING 172.31.8.16 (172.31.8.16) 56(84) bytes of data:
64 bytes from 172.31.8.16: icmp_seq=1 ttl=126 time=2.94 ms
64 bytes from 172.31.8.16: icmp_seq=2 ttl=126 time=1.14 ms
64 bytes from 172.31.8.16: icmp_seq=3 ttl=126 time=1.16 ms
64 bytes from 172.31.8.16: icmp_seq=4 ttl=126 time=1.16 ms
64 bytes from 172.31.8.16: icmp_seq=5 ttl=126 time=1.17 ms
64 bytes from 172.31.8.16: icmp_seq=6 ttl=126 time=1.14 ms
64 bytes from 172.31.8.16: icmp_seq=7 ttl=126 time=1.16 ms
64 bytes from 172.31.8.16: icmp_seq=8 ttl=126 time=1.19 ms
64 bytes from 172.31.8.16: icmp_seq=9 ttl=126 time=1.21 ms
64 bytes from 172.31.8.16: icmp_seq=10 ttl=126 time=1.17 ms
64 bytes from 172.31.8.16: icmp_seq=11 ttl=126 time=1.22 ms
64 bytes from 172.31.8.16: icmp_seq=12 ttl=126 time=1.16 ms
64 bytes from 172.31.8.16: icmp_seq=13 ttl=126 time=1.23 ms
64 bytes from 172.31.8.16: icmp_seq=14 ttl=126 time=1.19 ms
```

i-03598a28165fc6089 (my-ec2-vp02)

PublicIPs: 32.198.21.206 PrivateIPs: 10.0.75.50

```
[5]+ Stopped ping 10.0.0.0
[ec2-user@ip-172-31-8-16 ~]$ ping 32.198.21.206
PING 32.198.21.206 (32.198.21.206) 56(84) bytes of data:
64 bytes from 32.198.21.206: icmp_seq=1 ttl=126 time=0.676 ms
64 bytes from 32.198.21.206: icmp_seq=2 ttl=126 time=0.685 ms
64 bytes from 32.198.21.206: icmp_seq=3 ttl=126 time=0.678 ms
^Z
[6]+ Stopped ping 32.198.21.206
[ec2-user@ip-172-31-8-16 ~]$ ping 10.0.75.50
PING 10.0.75.50 (10.0.75.50) 56(84) bytes of data:
64 bytes from 10.0.75.50: icmp_seq=1 ttl=126 time=4.43 ms
64 bytes from 10.0.75.50: icmp_seq=2 ttl=126 time=1.29 ms
64 bytes from 10.0.75.50: icmp_seq=3 ttl=126 time=1.28 ms
64 bytes from 10.0.75.50: icmp_seq=4 ttl=126 time=1.28 ms
64 bytes from 10.0.75.50: icmp_seq=5 ttl=126 time=1.37 ms
64 bytes from 10.0.75.50: icmp_seq=6 ttl=126 time=1.27 ms
64 bytes from 10.0.75.50: icmp_seq=7 ttl=126 time=1.35 ms
64 bytes from 10.0.75.50: icmp_seq=8 ttl=126 time=1.29 ms
64 bytes from 10.0.75.50: icmp_seq=9 ttl=126 time=1.27 ms
^Z
[7]+ Stopped ping 10.0.75.50
[ec2-user@ip-172-31-8-16 ~]$
```

i-0ab13f9596bf87875 (public-ec2)

PublicIPs: 184.193.28.209 PrivateIPs: 172.31.8.16

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2. AWS VPC Endpoint

Objective

Configure VPC endpoints to securely access AWS services from a VPC **without requiring an Internet Gateway or NAT Gateway**, reducing exposure to the public internet.

Steps Performed

1. Selected the required VPC.
2. Created a **VPC Endpoint**.
3. Selected the AWS service that needs private connectivity.
4. For an S3/DynamoDB endpoint, configured a **Gateway Endpoint**.
5. Selected the required VPC route table(s).
6. Configured the endpoint policy according to the required access.
7. Verified that resources in the private subnet can access the AWS service through the endpoint.

```
[7]+ Stopped ping 52.72.87.241
[ec2-user@ip-172-31-72-14 ~]$ aws s3 ls
2026-04-19 10:25:14 zeeshan.agencypro
2026-05-16 10:13:42 zeeshanagency
2026-07-25 10:35:50 zeeshancloudtech.online
[ec2-user@ip-172-31-72-14 ~]$
```